

A Post-Quantum Secure Frequency-Domain Steganographic Framework Using Hybrid Encryption and Multi-Channel Embedding.

Nischal V Pattedar

*Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India
bl.en.u4aie23021@bl.students.amrita.edu*

Punav Anirudh Potluri

*Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India
bl.en.u4aie23025@bl.students.amrita.edu*

Raunak Choudhury

*Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India
bl.en.u4aie23027@bl.students.amrita.edu*

Gokul Shyam Lankoti

*Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India
bl.en.u4aie23015@bl.students.amrita.edu*

Abstract—The development of covert and quantum-resilient communication frameworks is driven by the growing viability of large-scale quantum computers, which presents a serious threat to traditional public-key cryptography systems. The design and implementation of a post-quantum safe image steganographic system that combines frequency-domain data concealing and hybrid encryption are presented in this study. In accordance with NIST post-quantum security guidelines, the suggested system protects session keys using the lattice-based CRYSTALS-Kyber768 key encapsulation mechanism while utilizing adaptive hybrid encryption with AES-256 and a lightweight polyalphabetic cipher for message confidentiality. A discrete cosine transform (DCT)-based frequency-domain embedding technique is used to deterministically partition and embed encrypted payloads across a carrier image's red, green, and blue spectral channels. Texture-adaptive masking and key-dependent sparse randomized sampling are used to maintain visual imperceptibility and improve resistance to statistical detection, limiting alterations to mid-frequency DCT coefficients and high-variance image regions. The entire system is developed as a working Java application that supports end-to-end secure communication and has both graphical and command-line interfaces. Reliable message recovery under common image handling operations and decreased detectability against a convolutional neural network-based steganalysis classifier trained using a ResNet-18 architecture which produced 98% accuracy in categorization of encrypted and normal images.

Index Terms—Post-quantum cryptography, image steganography, frequency-domain embedding, CRYSTALS-Kyber, hybrid encryption, covert communication.

quantum cryptography tackles confidentiality at the mathematical level, it does not address traffic analysis or the discovery of sensitive communication, which are still major issues in settings where monitoring is common. By hiding the actual existence of communication, steganography provides an extra degree of protection. However, with the use of deep learning-based detectors that can spot minute statistical aberrations, traditional spatial-domain steganographic methods are becoming more vulnerable to contemporary steganalysis. This work suggests a secure communication framework that combines frequency-domain image steganography with post-quantum cryptographic security to overcome these difficulties. The system uses the lattice-based CRYSTALS-Kyber768 key encapsulation method to protect session keys from quantum adversaries and a hybrid encryption strategy to safeguard message content. Using a discrete cosine transform-based method, encrypted payloads are inserted in the frequency domain of digital images, dispersing data across several color channels to lower statistical detectability. The embedding technique uses key-dependent sparse randomized sampling to distribute changes in a non-linear and predictable manner and texture-adaptive masking to avoid smooth image regions in order to maintain visual quality and improve robustness. The entire architecture is put into practice as a working software solution which is an end-to-end secure communication.

I. INTRODUCTION

The security of popular public-key cryptographic techniques like RSA and elliptic curve encryption is fundamentally threatened by the quick development of quantum computing. These systems could become computationally vulnerable due to algorithms like Shor's algorithm, which is driving the shift to post-quantum cryptography primitives that are safe from adversaries with quantum capabilities. Although post-

II. LITERATURE SURVEY

Achieving an ideal balance between computing efficiency, resilience against sophisticated adversaries, and preservation of perceptual quality is becoming more and more important in recent developments in multimedia security. Conventional encryption and steganography methods are becoming inadequate due to the quick adoption of deep learning-based analysis tools and the impending threat of quantum computing. As a

result, current research focuses on generative steganography, adversarial resilience, content-aware security, and hybrid cryptographic systems.

A. Content-Aware and Adaptive Encryption

Conventional selective video encryption results in predictable statistical distortions and needless processing complexity by applying uniform protection across frames. Sheng et al. [1] suggested a content-aware selective encryption scheme for H.265/HEVC that gives semantically sensitive segments priority in order to address this. In order to identify and selectively encrypt sensitive GOPs, such as those including humans, while preserving non-sensitive content, their method uses a deep hashing network based on Central Similarity Quantization. Hyper-chaotic Lorenz systems and coefficient-level obfuscation further improve security. Semantic awareness dramatically increases efficiency while preserving resilience against format-compliance and cryptanalytic assaults, according to experimental data, underscoring the increasing significance of adaptive, intelligent encryption over static techniques.

B. Robust and Adversarial Image Steganography

Covert communication is still susceptible to contemporary steganalysis, especially CNN-based detectors, even though adaptive encryption increases efficiency. In response, Wang et al. [2] developed an adversarial picture steganography technique that preserves local correlations and natural image statistics by clustering pixel modification directions using gradient feedback from deep-learning steganalyzers. Their method works better than previous adversarial embedding methods against both conventional and CNN-based detectors. In addition, Zhang et al. [6] studied batch image steganography for online social networks and presented the Multi-Stega framework, which improves robustness under lossy compression by reducing cumulative embedding artifacts across multiple JPEG images using embedding sign metrics and dynamic payload allocation. In order to stabilize quantized coefficients and achieve lower bit error rates while keeping good anti-steganalysis performance, Huang et al. [10] further addressed JPEG recompression robustness utilizing DCT Residual Modulation in conjunction with Reed–Solomon coding. When taken as a whole, these studies demonstrate a move toward channel-aware, statistically optimal steganography.

C. Lightweight and Contrast-Based Steganography

Despite their efficiency, deep learning-based algorithms have a significant computational cost that limits their practical application. To address this issue, Jamatia and Bhuyan [8] suggested a contrast-channel embedding method that uses deterministic contrast analysis instead of learning-based models. Their strategy improves imperceptibility while remaining simple by selecting embedding data in mid-range and severe contrast regions using dual-thresholding. This lightweight architecture provides a feasible alternative for real-time and resource-constrained applications, but its adaptability to advanced detectors is limited.

D. Generative and Provably Secure Steganography

Moving beyond modification-based approaches, Wang et al. [3] proposed SparSamp, a secure steganographic system that embeds messages during the construction of deep generative models. By utilizing sparse sampling without changing the underlying probability distribution, the approach achieves stringent theoretical security guarantees while preserving fast embedding speed. Similarly, Qasaimeh et al. [9] introduced a coverless steganography strategy based on GANs and compressive autoencoders, which dramatically increased embedding capacity and robustness. However, generative approaches frequently have large processing needs and sophisticated inversion operations, which limits their scalability.

E. Steganalysis and Adversarial Vulnerabilities

Steganographic techniques evolve, and so do detection strategies. Wei et al. [11] presented CTNet, a hybrid CNN-Transformer architecture that captures both local and global embedding artifacts, resulting in state-of-the-art detection accuracy for color image steganography. Kim et al. [12] showed that even advanced steganalysis algorithms are vulnerable to adversarial cases, highlighting significant trade-offs between detection accuracy and visual imperceptibility. These findings point to an increasing arms race between data concealing and detecting technologies.

F. Cryptographic Robustness and Post-Quantum Security

Beyond steganography, microarchitectural side channels and quantum computing are posing new challenges to cryptographic resilience. Barthe et al. [4] discovered that future processor improvements may cause significant leakage even in constant-time encryption schemes. Similarly, Du et al. [13] showed effective side-channel attacks on masked CRYSTALS-Kyber implementations, highlighting the vulnerability of “pure” post-quantum deployments. Hybrid cryptographic frameworks have grown in popularity as a means of improving efficiency and resilience. Selvi and Sakthivel [14] demonstrated the efficacy of ECC-AES hybrid encryption for cloud security, whereas Hanna et al. [15] conducted comprehensive benchmarking of PQC algorithms on IoT devices, highlighting the importance of selected and optimized deployment. Recent research have extended this paradigm by investigating lattice-based cryptography for safe multi-cloud storage [16] and evaluating PQC performance across heterogeneous environments [17], highlighting the importance of cryptographic adaptability.

G. Secret Sharing and Distributed Security

Distributed security mechanisms improve fault tolerance. Xiong et al. [7] introduced a secure secret image sharing technique based on polynomial k-consistency that protects against noise, manipulation, and cropping attacks. Similarly, JPEG-integrated secret sharing techniques [18] ensure format compliance while reducing key management issues. These ideas highlight the benefits of distributed systems in reducing single points of failure.

H. Entropy-Optimized Encryption and the Emergence of Post-Quantum Cryptography

Current research emphasizes the long-term effects of quantum computing as well as recent developments in cryptography optimization. In order to optimize Shannon entropy and get closer to the optimal value of 8, Stoycheva et al. [19] suggested an entropy-driven chaotic picture encryption strategy that adjusts third-order Lorenz system parameters. They demonstrate the efficacy of intelligent optimization in non-linear cryptographic systems by achieving near-perfect entropy, reduced pixel correlation, and faster convergence by including AI-assisted initialization into a modified Price's algorithm. Gitonga et al. [20], on the other hand, examined the systemic effects of quantum computing and shown through simulation that RSA-2048 and ECC-256 are extremely susceptible to Shor's method. According to their analysis, hybrid classical-post-quantum schemes are a viable step toward quantum-resistant cryptography and CRYSTALS-Kyber is a strong post-quantum option. Three main research gaps are identified by the evaluated literature. First, content sensitivity and entropy change are ignored by static, non-semantic encryption and embedding techniques, which results in inefficiency and observable patterns. Second, monolithic steganographic techniques lack resistance under partial compromise and are susceptible to contemporary Transformer-based detectors. Third, even though post-quantum cryptography is crucial, current implementations have performance issues and side-channel exposure, which emphasizes the necessity for hybrid and adaptable systems. An integrated architecture that combines lightweight adaptive steganography, cryptographically agile post-quantum security, and semantic-aware encryption is needed to close these gaps.

III. METHODOLOGY

In order to provide secure communication, the suggested system uses a layered approach that combines cryptographic security using quantum-resistant cryptographic algorithms with frequency domain steganography for image security. Cryptographic preparation, payload orchestration, frequency domain embedding, and recovery comprise the entire procedure.

A. Entropy-Guided Hybrid Encryption

Prior to steganographic embedding, the plaintext message undergoes hybrid encryption to ensure confidentiality against both classical and quantum adversaries. Let the plaintext message be denoted as:

$$M = \{w_1, w_2, \dots, w_n\},$$

where each w_i represents a word-level token. For each token, Shannon entropy is computed as

$$H(w_i) = - \sum_{c \in \Sigma} p(c) \log_2 p(c),$$

where $p(c)$ is the empirical probability of character c occurring in w_i .

An entropy threshold τ_e governs the encryption strategy:

$$E(w_i) = \begin{cases} \text{AES-256}(w_i), & H(w_i) \geq \tau_e, \\ \text{Vigenère}(w_i), & H(w_i) < \tau_e. \end{cases}$$

Encrypted tokens are concatenated to form the ciphertext C . The symmetric session key is encapsulated using the CRYSTALS-Kyber768 post-quantum key encapsulation mechanism, producing a protected payload P that is resilient to quantum adversaries.

B. Payload Partitioning and Channel Allocation

Following encryption, the ciphertext is deterministically separated into many pieces. In multi-channel payload distribution, the segments are subsequently assigned to the carrier pictures' red, green, and blue color channels. Consequently, the multi-channel strategy avoids concentrated changes in the payload distribution. Furthermore, channel allocation and the division of the ciphertext into many segments are carried out using logic that depends on the keys, which can only be recreated at the intended recipient.

C. Frequency-Domain Steganographic Embedding

The discrete cosine transform, or DCT, is used to embed data in the frequency domain for higher values of both imperceptibility and resilience. After segmenting the carrier picture into 8x8 non-overlapping blocks, it is converted into the frequency domain and data embedding is performed in the mid-frequency DCT coefficients. In order to permit embedding just in textured regions while disregarding smoother parts, which are more susceptible to attacks, texture adaptive masking computes block variance levels. Sparse randomized sampling based on the key is carried out in a non-linear deterministic manner to identify regions that are suitable for embedding in order to further guarantee embedding security. To guarantee simple retrieval without revealing any information about the embedding, the embedded message is injected using a parity modulation quantization technique with a constant persistence value.

D. Steganographic Image Transmission

The encrypted message is embedded over several spectral channels through steganography's dual use of error correcting codes. Without raising any red flags, the generated steganographic image can be stored or sent over traditional means. The cover image and the steganographic image continue to seem similar.

E. Payload Extraction and Message Recovery

At the receiver, the steganographic image I_s undergoes an inverse extraction pipeline. The image is decomposed into blocks, transformed using DCT, and candidate blocks are identified using the same variance threshold τ_v and key-dependent sampling sequence.

Payload bits are recovered as

$$b = \begin{cases} 1, & (D'_k \bmod 2) = 1, \\ 0, & (D'_k \bmod 2) = 0. \end{cases}$$

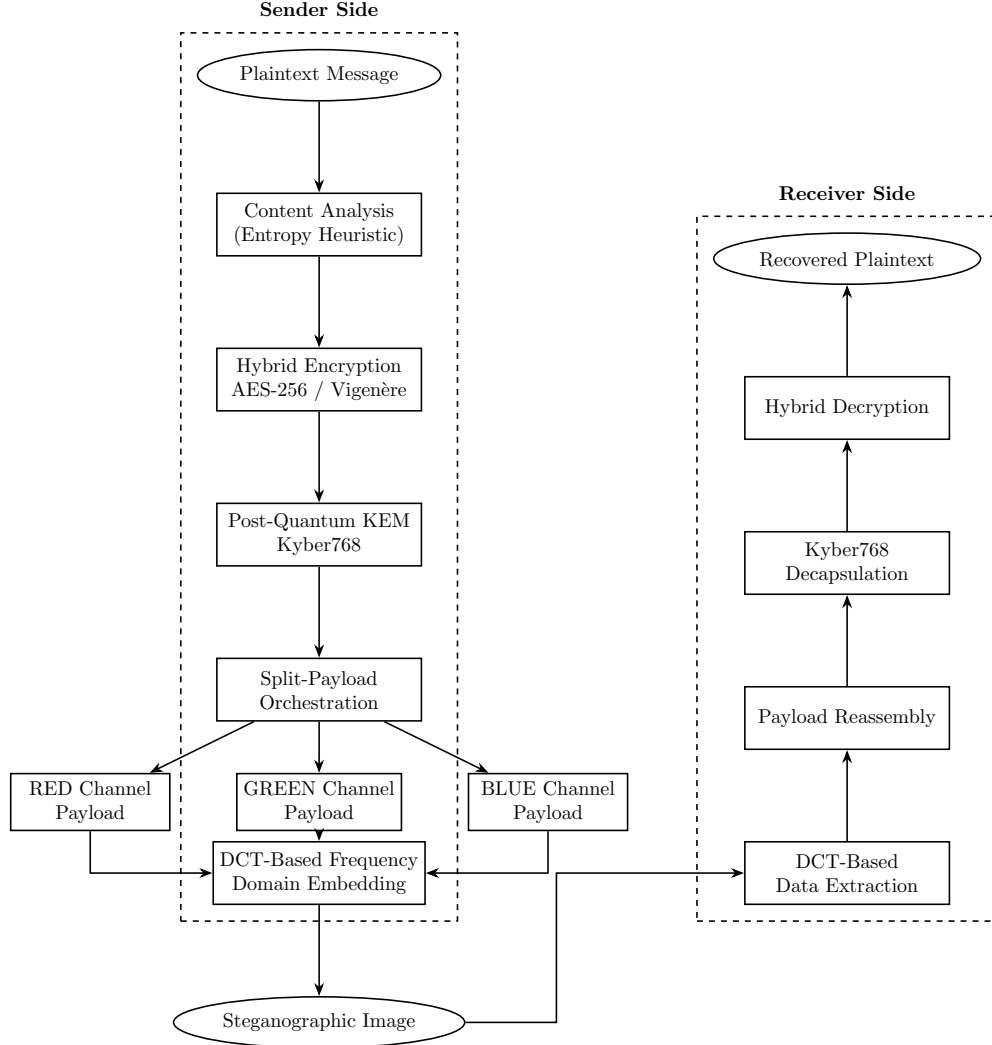


Fig. 1: System architecture of the post-quantum secure frequency-domain steganographic framework.

Extracted payload segments $\{P_R, P_G, P_B\}$ are reassembled to reconstruct P . The session key is decapsulated using Kyber768, followed by inverse hybrid decryption to recover the original plaintext message M .

F. Evaluation Methodology

The system is checked for imperceptibility, steganalysis resistance, and functional correctness. The successful recovery of the embedded messages confirms the accuracy. Visual inspection is carried out to verify that the perceptual impairment is negligible. The CNN-based steganalysis classifier with ResNet-18 architecture is trained on both the cover and stego pictures in order to assess the detectability.

The design of the suggested post-quantum secure steganographic framework is shown in Fig. 1. The encrypted payload is divided and embedded across several color channels of a carrier image using a DCT-based frequency-domain process with texture-adaptive masking and key-dependent randomized

sampling. On sender's side, the plaintext data is protected using hybrid encryption with session keys secured via the CRYSTALS-Kyber768 post-quantum key encapsulation mechanism. The original message at the receiver is recovered by inverse frequency-domain extraction, payload reassembly, post-quantum key decapsulation, and hybrid decryption.

IV. RESULTS AND DISCUSSION

The suggested framework was assessed in terms of resistance to machine-learning-based steganalysis, end-to-end steganographic communication, and cryptographic correctness. The completely implemented system and associated experimental evaluation are the sources of all the results presented in this section.

A. Cryptographic Correctness and Key Management

The effective production and administration of cryptographic keys using the CRYSTALS-Kyber768 key encapsulation technique confirmed the accuracy of the post-quantum

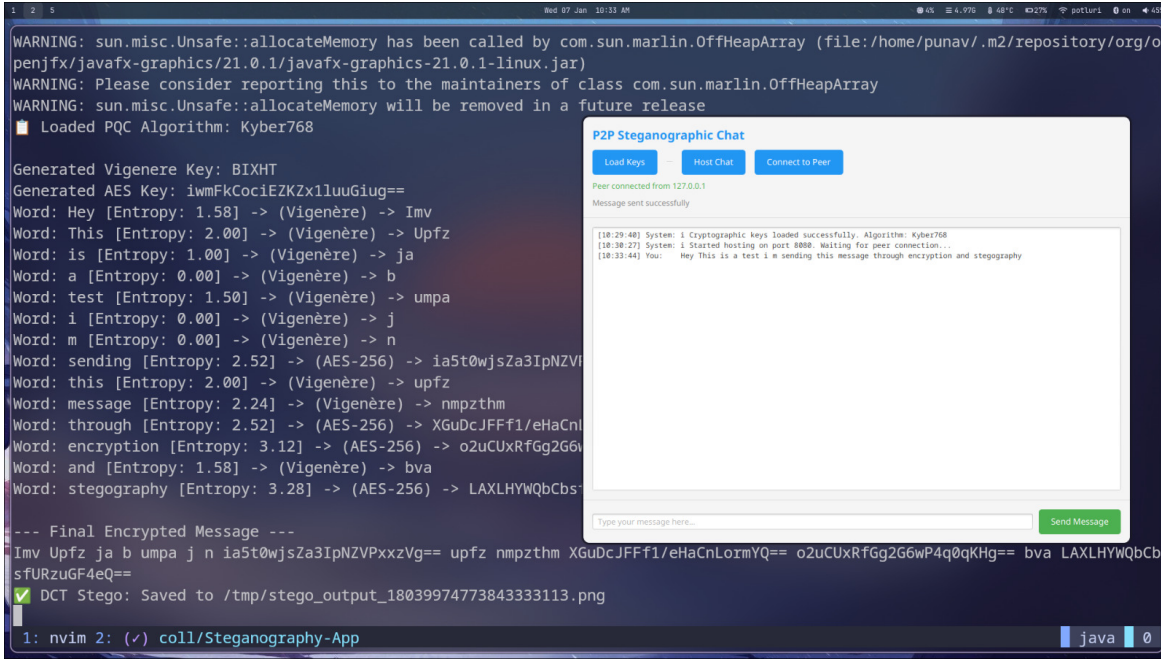


Fig. 2: End-to-end secure communication demonstrating entropy-driven hybrid encryption, DCT-based frequency-domain steganographic embedding, and successful peer-to-peer transmission of the steganographic payload.

cryptographic layer. The system confirms proper initialization of the hybrid cryptographic pipeline by generating symmetric AES and polyalphabetic Vigenère keys in addition to Kyber public-private key pairs, as illustrated in Fig. 3. The framework is appropriately set up for quantum-resilient key exchange, as seen by the presented key parameters and reported security level, which show conformity with NIST Level 3 post-quantum security standards.

B. End-to-End Secure Communication and Steganographic Embedding

Peer-to-peer communication, steganographic embedding, and encryption were successfully integrated, according to end-to-end testing. Entropy-driven hybrid encryption was used to process plaintext input during message transmission, producing a composite encrypted payload. Discrete cosine transform-based frequency-domain steganography was used to successfully embed the encrypted data into a carrier image, as shown in Fig. 2, and the steganographic image was produced error-free. System logs validated the whole operational pipeline from message input to secure delivery by confirming that the embedded payload was successfully transmitted via the peer-to-peer communication channel.

C. Evaluation Against CNN-Based Steganalysis

A convolutional neural network-based steganalysis model with a ResNet-18 architecture was trained to distinguish cover and steganographic images produced by the suggested method in order to assess resistance against automated detection. A purposefully unbalanced class distribution was reflected in the assessment dataset, which included 500 steganographic images

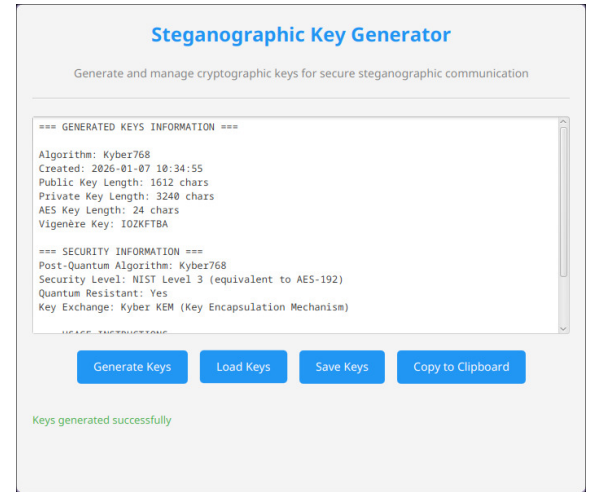


Fig. 3: Post-quantum key generation and hybrid cryptographic setup showing CRYSTALS-Kyber768 key encapsulation along with symmetric AES and Vigenère key material.

and 1000 cover images. This design decision avoids unduly optimistic performance estimations that could result from artificially balanced datasets and mimics realistic deployment circumstances where steganographic content is intrinsically scarce in comparison to benign media. The resulting confusion matrix is shown in Fig. 4. The classifier achieved an overall accuracy of approximately 98%, with only 4 cover images (0.4%) falsely classified as steganographic, indicating a very low false-positive rate. For steganographic images, 475 out of 500 samples were correctly identified, corresponding to a

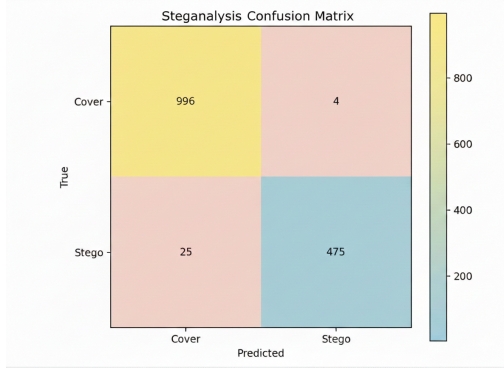


Fig. 4: Confusion matrix obtained from ResNet-18-based steganalysis model trained to classify original and steganographic images generated by the proposed method.

TABLE I: Steganalysis classification performance of the ResNet-18 model on cover and steganographic images.

Class	Precision	Recall	F1-score	Support
Cover (Original)	0.98	1.00	0.99	1000
Stego (Encrypted)	0.99	0.95	0.97	500
Overall Accuracy		0.98		1500

recall of approximately 95%. Quantitative performance metrics, including precision, recall, and F1-score for both classes, are summarized in Table I. Stratified train-test splitting was employed to preserve class proportions across all evaluation subsets, ensuring that reported metrics accurately reflect performance under imbalanced conditions.

The suggested method offers efficient defense-in-depth for covert communication in post-quantum threat environments, as demonstrated by the observed cryptographic correctness (Fig. 3), successful end-to-end operation (Fig. 2), and measured resistance to deep learning-based steganalysis (Fig. 4 and Table I). While formal undetectability guarantees are not claimed, the results confirm the suitability of the system for practical deployment scenarios requiring both quantum resistance and covert data transmission.

V. CONCLUSION

A post-quantum secure image steganographic system that combines hybrid encryption with frequency-domain data concealment was developed and empirically verified in this work. The CRYSTALS-Kyber768 post-quantum key encapsulation technology and symmetric encryption are successfully used in the suggested system to provide quantum-resilient key management while preserving practical deployability. All examined carrier images had visually undetectable data concealing thanks to frequency-domain embedding employing discrete cosine transform coefficients, multi-channel payload distribution, and texture-adaptive masking. The functional accuracy of the encryption, embedding, and extraction pipeline was confirmed by end-to-end evaluation, which showed successful message recovery across all experimental trials when correct crypto-

graphic keys were supplied. A ResNet-18-based steganalysis model was used to assess resistance to automated detection; it produced an overall classification accuracy of almost 98% on cover and steganographic images. The confusion matrix analysis showed a low false-positive rate (0.4%) for cover images and a false-negative rate of roughly 5% of overall steganographic images. This suggests that the suggested frequency-domain, multi-channel embedding strategy makes reliable detection more challenging, even for deep learning-based models. Collectively, our findings show that by addressing post-quantum resilience, cryptographic security, and covert communication, the suggested framework offers an efficient defense-in-depth solution.

VI. FUTURE SCOPE

Future improvements to the suggested system will concentrate on increasing assessment depth, enhancing steganographic robustness, and fortifying cryptographic assurances. By incorporating post-quantum digital signature techniques like CRYSTALS-Dilithium, the existing key exchange mechanism can be expanded at the cryptographic layer to allow for message-level integrity verification and authorized key exchange. The post-quantum key management module's implementation of forward secrecy and periodic session key rotation would significantly reduce exposure during extended communication sessions.

Future steganographic research will compare the robustness of various embedding domains, such as discrete wavelet transform (DWT) and perceptually uniform color spaces (like YCbCr), against picture alterations and compression. Learning-assisted optimization, in which lightweight models dynamically choose embedding locations based on local picture characteristics and detectability feedback, can supplement the existing rule-based, texture-adaptive coefficient selection. To strike a compromise between resilience and imperceptibility, payload capacity customization based on image content and channel circumstances can also be used. In terms of evaluation, the framework can be expanded to incorporate methodical testing under various payload sizes and post-processing attacks against sophisticated steganalysis models, such as transformer-based designs and cross-domain detectors. Lastly, using the system in high-throughput covert communication scenarios while maintaining post-quantum security assurances would be made possible by expanding it to support video and audio carriers as well as real-time embedding and extraction pipelines.

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